

RedHill Biopharma

GI-focused pharma with multiple catalysts ahead

Initiation of coverage

Pharma & biotech

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Price* **NIS4.4**

Market cap **NIS560m**

*Priced at 22 November 2016
NIS3.88:US\$

Net cash (\$m) at end Q316 40.5

Shares in issue 127.2m

Free float 88.0%

Code RDHL

Primary exchange TASE

Secondary exchange NASDAQ

Share price performance



% 1m 3m 12m

Abs (17.2) (24.2) (10.5)

Rel (local) (18.3) (22.7) (3.9)

52-week high/low NIS6.0 NIS3.3

Business description

RedHill Biopharma is an Israeli-based pharma company with a broad R&D pipeline focusing on inflammatory and gastrointestinal diseases, while earlier stage assets also target various cancers. The most advanced products are RHB-105 for *H. pylori* infection, RHB-104 for Crohn's disease and multiple sclerosis and Bekinda for gastroenteritis and IBS-D.

Next events

RHB-105 confirmatory Phase III start H117

1st DSMB review of RHB-104 Ph III in CD Q416

Bekinda Phase III for gastroenteritis top line results Mid-2017

Bekinda IBS-D Phase II top line results Mid-2017

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RedHill Biopharma is a biopharmaceutical company with a diversified, balanced R&D pipeline and late-stage assets focusing on gastrointestinal (GI) diseases and inflammation. The three most advanced Phase III-stage products have been significantly de-risked, target areas with high unmet need and may deliver four key readouts with the next 18 months. We initiate coverage of RedHill with a valuation of NIS1.21-1.29bn or NIS9.5-10.1/share.

Year end	Revenue (\$m)	PBT* (\$m)	EPS* (\$)	DPS (\$)	P/E (x)	Yield (%)
12/14	7.0	(10.7)	(0.12)	0.0	N/A	N/A
12/15	0.0	(21.1)	(0.19)	0.0	N/A	N/A
12/16e	0.0	(25.1)	(0.20)	0.0	N/A	N/A
12/17e	0.1	(32.9)	(0.26)	0.0	N/A	N/A

Note: *PBT and EPS are normalised, excluding amortisation of acquired intangibles, exceptional items and share-based payments.

Key late stage assets to deliver multiple catalysts

RedHill's main value drivers are late-stage assets in GI and the inflammation area, while in total the company has eight clinical-stage products. RedHill's strategy focuses on both established and innovative technologies, acquiring them with low upfronts and leveraging the potential by expanding the label, formulating novel combinations or targeting indications that are not too crowded with branded drugs. The three most advanced assets include RHB-105 for *Helicobacter pylori* (*H. Pylori*) infection (confirmatory Phase III to start in H117); RHB-104 for Crohn's disease (CD) (first DSMB in Q416, second DSMB expected in Q217 and to include an option for early termination) and relapsing-remitting multiple sclerosis (final results Q416); and Bekinda for gastroenteritis (top-line results expected in mid-2017) and diarrhoea-predominant irritable bowel syndrome (IBS-D; top line results mid-2017).

Focus on GI and inflammation

RHB-105 is a patented combination of rifabutin, amoxicillin and omeprazole for *H. pylori* infection regardless of ulcer status, an indication potentially broader than competitors. RHB-104 is a combination of clarithromycin, cefazolin and rifabutin for CD patients and represents a novel approach aimed at treating *Mycobacterium avium paratuberculosis* (MAP) infection, which according to numerous studies can be one of the causative agents of CD. Bekinda is an extended release, orally administered ondansetron in Phase III for gastroenteritis and in Phase II for IBS-D.

Valuation: NIS1.21-1.29bn or NIS9.5-10.1/share

We initiate coverage of RedHill with a valuation of NIS1.21-1.29bn or NIS9.5-10.1/share, based on risk-adjusted NPV analysis and including \$40.5m net cash at end Q316. In our valuation we include the company's five most advanced assets of its eight for a total of nine indications. The three most advanced assets constitute 58% of our valuation; all are in Phase II or III and may deliver four key readouts within next 18 months. In addition, any potential licensing deals could be supportive of the share price.

Investment summary

Company description: Diversified pharma with GI focus

RedHill Biopharma, an emerging Israel-based biopharmaceutical company, is focused on acquiring, developing and commercialising (in partnerships or alone) advanced clinical-stage, patent-protected, orally administered, small molecule drugs for the treatment of inflammatory and GI diseases, while earlier stage assets also target various cancers. In line with this strategy, RedHill in-licensed or purchased all of its products from different partners (more detail in Exhibits 1 and 7) and now the R&D pipeline includes eight clinical-stage products. RedHill listed on the Tel Aviv stock exchange in February 2011, raising \$13.6m, then carried out a dual listing on Nasdaq in 2012, raising \$6.5m. Since then Redhill has completed a number of secondary offerings adding \$80m.

Valuation: NIS1.21-1.29bn or NIS9.5-10.1/share

We value RedHill at NIS1.21-1.29bn or NIS9.5-10.1/share, based on risk-adjusted NPV analysis using a 12.5% discount rate. Our valuation includes \$40.5m net cash and other short-term financial assets at end Q316 and is based on the 52-week NIS exchange rate range to the US dollar of 3.74-3.99. According to current R&D plans, we expect trial readouts in 2017 for key products, which could trigger out-licensing deals (Exhibit 1). In our valuation we include five of the company's eight assets and in a total of nine indications. The inclusion is mostly based on the stage of the development and clear plans for progress. The company's strategy is to develop the products standalone or out-license in later stages. Among the products in our valuation only Rizaport's rights were licensed out in one territory so far, while other products are still wholly owned. We assume that the development of the products in Phase III will be finalised by RedHill, but a commercial partner or distributor will be sought due to the number of specialty areas in which the company is involved.

Financials: Cash reach beyond catalysts in 2017

RedHill had cash and cash equivalents (including bank deposits and financial assets at fair value) of \$40.5m at the end of Q316 vs \$64.2m in Q315 and was debt free. We expect a cash position of \$33.2m by the end of this year, which should provide a cash reach into Q417, well past several key R&D catalysts. Notably, we do not take into account potential revenues from Rizaport (although it is included in our valuation) or other products and we do not include any potential licensing-related income in our financial forecasts. RedHill may also attract additional non-dilutive funding via grants as it has in the past.

Sensitivities: Diversified R&D pipeline

RedHill is subject to the usual risks associated with drug development, including clinical development delays or failures, regulatory risks, competitor successes, partnering setbacks and financing and commercial risks. Some of the company's lead products (eg RHB-105, RHB-104 and Bekinda) are repositioned or reformulated generic drugs, which implies a lower R&D risk. On the other hand, if approved they may face generic substitution risk. The competitive advantage of these products versus cheaper generics will dictate the uptake of the market. Also, the composition of matter patents have already expired for the aforementioned products, hence making the overall IP protection vulnerable to challenges; however, formulation patents may provide adequate protection. The biggest near-term sensitivities are related to RHB-105, RHB-104 and Bekinda, which are all in late-stage development. There is also substantial uncertainty about the marketing of the products as they all target rather large patient populations, so a strong marketing partner is likely to be needed. According to our model, the company will need to carry out a fund-raising during the next 12 months.

Outlook: Broad R&D pipeline and eventful 2017

RedHill's R&D portfolio is rather diverse in terms of the products' mechanism of action and therapeutic areas. Exhibit 1 summarises the assets, the indications each product is being developed for and upcoming events (see Exhibit 7 for in-licensing deal details). The most advanced products are **RHB-105**, which is being developed for *H. pylori* (second Phase III to be initiated in H117), **RHB-104** for CD (enrolment to be completed by end 2017) and MS (final results in Q416), and **Bekinda** for gastroenteritis (top line results expected mid-2017) and IBS-D (top line results mid-2017).

Exhibit 1: RedHill's R&D pipeline, current status and upcoming newsflow

Product	Stage	Indication	Comments and upcoming events
RHB-105 rifabutin+ amoxicillin+ omeprazole	Ph III	<i>H. pylori</i> infection	In-licensed from Giaconda in August 2010. RHB-105 is being developed as a first-line treatment for patients diagnosed with <i>H. pylori</i> infection regardless of ulcer status – "test and treat" strategy, which is gaining traction. RedHill completed the first Phase III. Following a pre-Phase III meeting with the FDA, a confirmatory Phase III is planned to be initiated in H117. Subject successful outcome the study is expected to allow for the NDA application.
RHB-104 clarithromycin+ clofazimine+ rifabutin	Ph III	Crohn's disease	In-licensed from Giaconda in August 2010. Interim data and safety monitoring board (DSMB) review in Q416. By October 2016, 219 out of planned 410 patients were enrolled. Enrolment should be completed by end of 2017 and we expect the results to be out late in 2018 or early 2019.
	Ph IIa	r/r multiple sclerosis	Top-line final results from Phase IIa CEASE-MS study for r/r MS with RHB-104 as an add-on therapy to interferon beta-1a are expected in Q416 . Top-line interim results in March 2016 demonstrated positive safety/efficacy signals.
Bekinda ext. release tab. ondansetron	Ph III	Gastro- enteritis	In-licensed from Temple University in the US. Top-line results expected in mid-2017.
	Ph II	IBS-D	Top line results from Phase II with IBS-D patients expected mid-2017.
Yeliva sphingosine kinase-2 inhibitor	Ph I/II	r/r multiple myeloma	In-licensed from Apogee in March 2013. Phase I/II initiated in September 2016.
	Ph II	HCC	Phase II initiated in October 2016.
	Ph I/II	DLBCL	Phase I/II study was initiated in June 2016. Currently on administrative hold to amend the protocol aimed at improving enrolment.
	Ph Ib	Radioprotectant	A Phase I/II study oral mucositis in radiotherapy patients is expected to be initiated later in 2016.
Rizaport oral thin film rizatriptan	Market	Migraine	Co-development agreement with IntelGenx in 2010. Re-submission of NDA expected in H117. Received MAA in Europe in October 2015. First distribution agreement in Spain signed with Grupo Juste in July 2016. Ongoing discussion with more potential commercialization partners.
Other opportunities			RHB-106 , capsules of sodium picosulphate for bowel preparation for abdominal procedures; licensed to Salix Pharmaceuticals in February 2014, which was acquired by Valeant Pharmaceuticals in March 2015. It has yet to clarify further development plans. Mesupron , urokinase-type plasminogen activator (uPA) inhibitor, for solid tumours; in-licensed from Willex in June 2014, which explored Mesupron in 10 clinical studies including two Phase II studies in advanced pancreatic cancer and metastatic breast cancer. RedHill is conducting additional preclinical studies to refine the development strategy. RP101 , heat shock protein 27 inhibitor, acquired option to in-license RP101 from RESprotect in August 2014, which explored RP101 several Phase I and II studies with pancreatic cancer. The exercise term extended to May 2017. RedHill is conducting additional preclinical studies.

Source: Edison Investment Research. Note: IBS-D: irritable bowel disease with diarrhoea; r/r: relapsing-remitting multiple sclerosis/refractory or relapsed multiple myeloma; DLBCL: diffuse large B-cell lymphoma; HCC: hepatocellular carcinoma

RHB-105 for *H. pylori* eradication

RedHill acquired RHB-105 from Sydney-based Giaconda in August 2010 (see Exhibit 7 for details, including royalty rates). RHB-105 is a proprietary fixed-dose oral combination of two antibiotics (rifabutin and amoxicillin) and a proton pump inhibitor (PPI, omeprazole) in development for the treatment of *H. pylori* infection. The prevalence of *H. pylori* infection is widespread, estimated at 30-40% in industrialized countries and 70% in developing countries ([Centers for Disease Control and Prevention, CDC](#)). It is strongly associated with a wide range of upper gastrointestinal diseases, including gastritis, peptic ulcer disease and gastric cancer. Treatment of *H. pylori* remains a challenge for physicians as there is no single effective regimen (Exhibit 2). Commonly recommended first-line therapy is a PPI-based triple therapy, however, in recent years eradication rates fell to around 70%, with increasingly more patients remaining *H. pylori* positive, mainly due to increasing antibiotic resistance to clarithromycin and/or metronidazole.¹ Alternative therapies, including bismuth-containing quadruple therapy, sequential, concomitant and hybrid therapies are used in areas of high

¹ P. Hsu. Randomized Controlled Trial Comparing 7-Day Triple, 10-Day Sequential, and 7-Day Concomitant Therapies for Helicobacter pylori Infection. Antimicrobial Agents and Chemotherapy, October 2014 Vol 58.

clarithromycin resistance or as second-line therapy after standard triple therapy failure ([Maastricht IV/Florence Consensus Report guidelines](#)). Nevertheless, these regimens fail to eradicate *H. pylori* in around 10% of all cases.² Rifabutin is a rifamycin-S derivative used to treat *Mycobacterium species* (specific primarily lung infection in immunocompromised HIV infected patients) and represents a potential strategy for *H. pylori* eradication as it does not share resistance to clarithromycin. The mean *H. pylori* rifabutin resistance rate was found to be only 1.3%.⁴ The antibiotics that constitute RHB-105 are currently an option for a rescue therapy if other regimens fail, although they are rarely used.⁴ However, RHB-105 contains doses that are different from those in available generics; therefore the PK/PD profile should be a differentiating aspect. RedHill targets first-line treatment indication, which if successful would significantly expand the use of rifabutin-based regimen.

Exhibit 2: Main regimens for *H. pylori* treatment; RHB-105 contains omeprazole + amoxicillin + rifabutin

Regimen	Components	Use	Eradication (%)	Disadvantages
Standard triple therapy (7-14 d)	PPI + clarithromycin + amoxicillin (or metronidazole)	First line in areas of low clarithromycin (CAM) resistance	70-85% (14% in resistant strains)	Widespread clarithromycin resistance
Bismuth quadruple therapy (10-14 d)	PPI + bismuth + tetracycline + metronidazole	First line in areas of > 20% CAM resistance. 2nd line when triple therapy fails.	75% - 90%	Bismuth has some toxicity issues; complex dosing, many tablets
Sequential therapy, (5d + 5d)	Dual therapy; PPI + amoxicillin followed by triple therapy; ppi + clarithromycin + metronidazole	As above	c. 70% in CAM resistant strains (40% in dual resistant strains)	Dual resistance problems
Levofloxacin-based triple therapy, (10d)	PPI + levofloxacin + amoxicillin	Second or third line. Broad-spectrum quinolone used to overcome CAM resistance.	c. 90%	Quinolone resistance increasing (40% in US, 20% Europe, 10% Asia)
Rifabutin-based triple therapy, (7-14d)	PPI + amoxicillin + rifabutin	'rescue' treatment	67-79%, Gisbert et al ⁴ 89.4%, RedHill data	Rifabutin has rare side effects (myelotoxicity, ocular toxicity), ⁴ although RedHill did not encounter significant side effects; also RHB-105 PK/PD profile is different as rifabutin dose is divided into three 50mg per day.

Source: adapted from V. Papastergiou et al. Treatment of Helicobacter pylori infection: Past, present and future. World J Gastrointest Pathophysiol. 2014 Nov 15; 5(4): 392-399. Note: PPI: Proton Pump Inhibitor (lansoprazole/ rabeprazole/ esomeprazole) at standard dose; CAM: clarithromycin

Initial clinical data encouraging

A Phase II proof-of-concept study was published in 2006 by Prof Borody et al., sponsored by Giaconda, validating the use of rifabutin in *H. pylori* eradication. Australian patients who had failed one or more clarithromycin-based treatments received triple therapy containing rifabutin, amoxicillin (low or high dose) plus pantoprazole for 12 days. Results were very encouraging with eradication rates of 90.8% in the low-dose amoxicillin group, and 96.6% in the high dose group (Exhibit 3).³

Exhibit 3: RHB-105 trials

Study	Design	Patients	Status	Outcome (eradication %)
Phase II proof of concept (Australia)	Two arms: triple therapy containing separate rifabutin (150mg daily), amoxicillin (1g OR 1.5g tid) and pantoprazole (80mg tid) for 12d	130 pts who had failed clarithromycin-based therapy; Rescue therapy	Reported 2006 (Borody et al, 2006)	Low amoxicillin arm: 90.8% High amoxicillin arm: 96.8% (not significantly different).
Phase III Eradicate Hp (US)	Double-blind. Active arm: RHB-105 containing rifabutin 150mg/ amoxicillin 3g/ omeprazole 120mg daily (4 capsules every 8 hours) for 14d. Control arm - no treatment as a control for primary endpoint, when unblinded received standard triple therapy.	118 pts with active dyspepsia and confirmed <i>H. pylori</i> ; First line therapy	Top line results Jun 2015; final results Mar 2016	Treatment arm: 89.4%, superior eradication over historical 70% levels (p<0.001). Control: 63% eradication with SOC after unblinding
Phase III confirmatory (US)	Phase III – confirmatory. Two arms: RHB-105 vs dual therapy (high dose amoxicillin/omeprazole)	Approx. 440 pts. Details TBD.	Planned start Q416/Q117	N/A

Source: RedHill, Edison Investment Research. Note: Tid – three times a day; SOC – standard of care

² T. Rokkas et al. Cumulative *H. pylori* eradication rates in clinical practice by adopting first and second-line regimens proposed by the Maastricht III consensus and a third-line empirical regimen. Am J Gastroenterol. 2009 Jan;104(1):21-5.

³ T. J. Borody. Efficacy and safety of rifabutin-containing 'rescue therapy' for resistant Helicobacter pylori infection. Aliment Pharmacol Ther. 2006 Feb 15;23(4):481-8.

RedHill carried out a Phase III study (ERADICATE Hp) in the US in patients with dyspepsia, looking at *H. pylori* eradication after 14 days of treatment with RHB-105 as a first-line therapy. The final results were published in March 2016 and confirmed an eradication rate of 89.4% for RHB-105. The trial met its primary endpoint of superiority over a 70% historical efficacy rate. The eradication rate in the placebo group with standard of care treatment was 63% (after unblinding, see Exhibit 3). No treatment-related serious adverse events were noted. Following a meeting with the FDA in April 2016, RedHill is planning a second confirmatory US Phase III study to start H117. This double blind, randomized trial will compare RHB-105 against a high dose amoxicillin/omeprazole, as the FDA agreed to a dual regimen instead of usual the triple due to prevalent resistance to clarithromycin.

Market opportunity and RHB-105 positioning

The Qualified Infectious Disease Product (QIDP) designation granted by the FDA in 2014 endows both fast-track development status and priority review status, and, if approved, eight years US market exclusivity. RHB-105's potential competitive advantages include superior efficacy, a broader indication and simpler treatment regimen with all-in-one capsules than that of generic peers, including that it has only two antibiotics, which may appeal to physicians and pharmacologists afraid of increasing overall resistance.

The planned indication is the first-line treatment for eradication of *H. pylori* regardless of ulcer status, which would be broader than its competitors: Prevpac (first-line regimen; lansoprazole, amoxicillin, clarithromycin; Takeda) and Pylera (second line regimen; bismuth, metronidazole and tetracycline; Actavis/Allergan; PPI has to be administered separately) are indicated for patients with *H. pylori* and duodenal ulcer disease (active or historic). A potential hurdle will be the threat of substitution with generics as all components in RHB-105 are available in a generic form; however the dose of rifabutin in RHB-105 is 50mg given three times a day, while the generic form is 150mg given once, which would not allow for exact substitution including different PK/PD profiles. One of the main competitive advantages will be a more convenient dosing compared to multi-pill generic combinations, which would increase compliance. The conventional regimens include two to three different antibiotics administered three to four times a day, with a proton pump inhibitor (eg omeprazole) two times a day. The RHB-105 treatment is four capsules three times daily, similar to other all-in-one therapies like Prevpac (four capsules twice daily) and Pylera (three capsules four times a day.) Prevpac sales peaked at \$150m in 2009 in the US (Bloomberg) before the patent expiry, while Pylera peak sales reached \$38m in the US in 2013. The retail price of branded version of Prevpac is \$950-1,000 per 14-day treatment in the US, while generic version retails for \$165-580 per 14-day treatment (GoodRx.com). Branded Pylera costs \$740-780 per 10-day treatment course in the US. Both combination products are composed of antibiotics used in first- and second-line regimens, hence they experienced a steady increase in resistance over the past decade. Currently generic rifabutin-based therapy (PPI+amoxicillin+rifabutin; RHB-105's constituents) has a retail price of around \$200 per 14-day treatment in the US but, since rifabutin-based therapy is not approved specifically for *H. pylori* eradication, it means off-label use.

Current guidelines discuss rifabutin as a rescue therapy when other second-/third-line options have failed.^{4,5} However, in our view, if the RHB-105 confirmatory Phase III study is supportive, this product will immediately be an attractive option for third-line or rescue treatments with the potential to move towards front line, once consensus emerges. For example, the Kyoto Global Consensus Meeting in 2015 achieved a high level of agreement on the need to distinguish between functional dyspepsia and

⁴ J. P. Gisbert, X. Calvet. Rifabutin in the Treatment of Refractory Helicobacter pylori Infection. Aliment Pharmacol Ther. 2012;35(2):209-221.

⁵ W. Chey. American College of Gastroenterology Guideline on the Management of Helicobacter pylori Infection. Am J Gastroenterol 2007;102:1808–1825.

H. pylori-associated dyspepsia and proposed diagnostic algorithms and eradication strategies in spite of the ulcer status.⁶

The US market potential alone is significant with over 100 million Americans thought to be *H. pylori*-positive and 500,000-850,000 new cases of peptic ulcer disease annually (CDC). Given the observations that around 10-15% (excluding gastroesophageal reflux disease) of the US population has dyspepsia and around 10% of those receive eradication treatment, this translates into c 2.9 million patients representing target population for RHB-105.^{7,8}

RHB 104 – new frontiers in autoimmune diseases

RHB-104 is a patented combination of three generic antibiotics (clarithromycin, rifabutin and clobazimine) in an oral capsule for the treatment of MAP infections and was also acquired by RedHill from Sydney-based Giaconda in August 2010. An increasing amount of data supports the link between MAP infection in CD patients and RedHill believes it could induce and prolong remission time by removing one of the exacerbating factors. RHB-104 is in Phase III development for CD, an area where current therapies have limited efficacy and pronounced side-effects, and are often very costly. The first DSMB interim analysis in the ongoing Phase III study with RHB-104 for CD is expected in Q416. A Phase IIa proof of concept study for multiple sclerosis (MS) is also ongoing; encouraging top-line interim results were announced in March 2016 and final results are expected Q416.

The MAP hypothesis and CD

Inflammatory bowel disease (IBD), which includes CD and ulcerative colitis (UC) is a lifelong condition with serious quality of life implications. CD is characterised by inflammation of the gastrointestinal (GI) tract, and symptoms include persistent diarrhoea, abdominal pain, rectal bleeding, weight loss and fatigue. Though studies strongly suggest an important genetic component, NOD2/CARD15 gene mutations, which influence immune response, are found in only around 20% of IBD patients (CCFA). It is proposed that a combination of genetic susceptibility and environmental factors may trigger an abnormal immune response (CCFA). One such factor is MAP, which is the causative agent of Johne's disease, a disease of cattle clinically and pathologically similar to CD, and is seven times more likely to be detected in CD patients compared to UC /non-IBD controls.⁹ Whether MAP could be a causal agent of CD remains in question, but the use of anti-mycobacterial antibiotics to treat CD has been shown in a number of clinical studies.⁹ Although a large Phase III trial funded by Pharmacia/Pfizer failed to meet its primary endpoints, several flaws in this trial have been highlighted (discussed below); the causality of MAP in CD is inconclusive and the need for a well-designed trial is compelling.¹⁰

Key clinical data on MAP hypothesis so far

Before RedHill, several clinical trials with RHB-104 were conducted with earlier formulations of the triple antibiotic combination, including two positive Phase II studies by Prof Borody associated with Giaconda (Exhibit 4) and the controversial Phase III by Pfizer. In 2000 Giaconda licensed the triple antibiotic to Pharmacia, which was acquired by Pfizer in 2003. The Phase III trial with 213 CD patients demonstrated significantly better short-term remission in the antibiotic arm than the placebo arm at 16 weeks.¹⁰ However, the authors concluded there was 'no evidence of sustained benefit' of the anti-mycobacterial treatment in CD as relapse rates were not statistically significantly different from week

⁶ K. Sugano. Kyoto global consensus report on *Helicobacter pylori* gastritis. Gut. 2015 Sep; 64(9): 1353–1367.

⁷ Y. Shaib et al. The prevalence and risk factors of functional dyspepsia in a multiethnic population in the United States. Am J Gastroenterol. 2004 Nov;99(11):2210-6.

⁸ C. Howden et al. Practice Patterns for Managing *Helicobacter pylori* Infection and Upper Gastrointestinal Symptoms. Am J Manag Care. 2007;13:37-44.

⁹ R. Chiodini et al. Crohn's disease and the mycobacterioses: A quarter century later. Causation or simple association? Critical Reviews in Microbiology, 2012; 38(1): 52–93.

¹⁰ W. Selby et al. Two-year combination antibiotic therapy with clarithromycin, rifabutin, and clobazimine for Crohn's disease. Gastroenterology. 2007 Jun;132(7):2313-9. Epub 2007 Mar 21.

104 onward. Pfizer subsequently discontinued development. However, flaws in the design of the trial received criticism;^{11,12} namely:

- the study failed to take into account the effect of the patients excluded from the study in the initial remission induction stage and chose relapse rates as primary endpoint, while remission may be a more representative endpoint;
- the study may have been underpowered since there is a wide reported range of MAP infections among CD patients, ranging from 50% to 90%;
- the study did not identify which patients were infected with MAP and effects of the treatment;
- the doses of antibiotics were much lower than in previous trials; and
- cefazolin bioavailability was hampered by a double capsule.

In their reanalysis of the Pfizer study Behr and Hanley suggests that looking at remission in the intention-to-treat population from week 16 instead of relapse rates would be more appropriate and would have shown statistically significant improvements in remission in the treatment arm over placebo. The original study design required a very high response rate for a positive outcome. RedHill's Phase III study uses a remission endpoint and is using optimised antibiotic doses.

Exhibit 4: Summary of RHB-104 CD clinical trials

Trial/Investigator	Design	Outcome	Trial
Phase II (2002) Borody (Giaconda)	Open label; 12 patients with severe CD, treated with triple antimycobacterial therapy*; 54 month follow-up.	Reversal of severe CD in 6/12 patients: complete clinical, colonoscopic and histologic remission of CD	Borody et al., 2002
Phase II (2005) Borody (Giaconda)	Open label; 52 patients with active CD, treated with triple antimycobacterial therapy** for between 6m-9yrs. Data reviewed retrospectively	CDAI reduction of >70 in 60% of patients (P<0.001) Baseline CDAI score had a significant effect between responders and non-responders (n=8) Marked improvements in mucosal healing, pain and diarrhoea	Borody et al., 2005 (abstract)
Phase III (2005) Pharmacia/Pfizer	Placebo controlled double blind, two-arm, prospective trial; 213 patients with CD, treated with triple antimycobacterial therapy*** or placebo, in addition to a 16-week tapering course of prednisolone. After week 16, subjects in remission (CDAI ≤150) were allowed to continue study medication. Primary endpoint was proportion of patients with at least one relapse at 12, 24 and 36 months	At week 16, remission higher in antibiotic arm than placebo (66% vs 50%, p = 0.02) By week 52, relapse rate was 39% on antibiotics vs 56% placebo (p=0.054) By week 104 relapse rate was 26% vs 43% (p = 0.14) and by 36 months, relapse rate was 59% vs 50%. Thus authors concluded 'no evidence of sustained benefit' - Pfizer discontinued development	Selby et al, 2007
US Phase III ongoing (started Sept 2013) RedHill	Placebo-controlled, double-blind, two-arm, prospective trial; will enrol 410 patients with moderate to severe CD. Treatment with daily RHB-104**** or placebo. Primary endpoint is % of patients in remission (CDAI ≤100) at 26 weeks. Secondary endpoints include response at week 26 (reduction in CDAI by 100 points), duration of remission/response and maintenance of remission weeks 26-52	In total three DSMB reviews are expected, with the first one planned in Q416. Second DSMB review is expected in Q217 and will include an option for early termination if overwhelming efficacy is demonstrated. If the trial is not terminated after the second DSMB review in Q217, the completion of recruitment is expected by the end of 2017.	[NCT01951326]
Europe Phase III	Planned	N/A	N/A

Source: Edison Investment Research. Note: CDAI – CD activity index ; *given separately: rifabutin (450mg/d), clarithromycin (750mg/d) and cefazolin (2mg/kg/d); **given separately: rifabutin (600mg/d), clarithromycin (1g/d) and cefazolin (100mg/d); ***given separately rifabutin (450mg/d), clarithromycin (750mg/d) and cefazolin (50mg/d); ****RHB-104 capsule: rifabutin (450mg/d), clarithromycin (950mg/d) and cefazolin (100 mg/d).

New Phase III trial of RHB-104 should answer MAP question

In September 2013, RedHill initiated a new [Phase III](#) trial (several design modifications were announced in October 2016) in moderate to severe CD (the MAP US study), conducted across 150 sites (Exhibit 4). By October 2016, 219 out of the planned 410 patients were enrolled and DSMB safety interim analysis is planned for Q416. Two additional DSMB reviews are planned, with the second one expected in Q217, which will also include interim efficacy analysis with an option for early termination if overwhelming efficacy is demonstrated. RedHill also expects that a second Phase III study in CD will be needed before the regulatory application. MAP infection is very difficult to detect in

¹¹ [Correspondence. Gastroenterology 2007;133:1742–1748.](#)

¹² M. Behr and J. Hanley. Antimycobacterial therapy for Crohn's disease: a reanalysis. The Lancet Infectious Diseases, volume 8, No. 6, p344, June 2008.

humans and as part of its strategy RedHill is developing, in collaboration with Quest Diagnostics, a commercial diagnostic test to detect MAP in blood. RedHill also collaborates with three universities in the US to develop PCR-based MAP companion diagnostic tests.

Crohn's disease – significant unmet need

According to the Crohn's & Colitis Foundation of America (CCFA), CD affects up to 700,000 Americans (UC is a similar number) with c 30k new cases every year, most of whom are diagnosed by the age of 35. In Europe there are estimated 1.6 million CD sufferers with 78k new cases every year.¹³ 45% of those in remission will remain relapse-free over the next year, while 35% will have one of two relapses (CCFA). There is no cure for CD, though a variety of immunomodulating and immunosuppressive agents are commonly used to control symptoms. Many of these are associated with significant failure rates, side effects and safety issues (J&J's Remicade has a black box warning on serious opportunistic infections like tuberculosis). Cost per patient ranges from \$300-400 per year for the cheaper generic aminosalicylates to in excess of \$20k per year for biologics (anti-TNFs such as AbbVie's Humira, UCB's Cimzia and J&J's Remicade and Simponi; or integrin-receptor antagonists such as Biogen's Tysabri and Takeda's Entyvio; source: GoodRX.com). The value of the global CD drug market is projected to be \$7.9bn by 2020 (EvaluatePharma).

RHB-104 in multiple sclerosis and other indications

In addition to CD, MAP has been associated with other diseases including type 1 diabetes, Hashimoto's thyroiditis, sarcoidosis and MS.¹⁴ MS is an inflammatory, neurodegenerative disease when a person's own immune system attacks the neurons in the central nervous system, leading to a variety of disabling symptoms that usually worsen over time. It has been proposed that a molecular mimicry between MAP proteins and human proteins could induce autoimmune pathologies and, in the case of MS, there could be similarities between MAP proteins and human anti-myelin basic protein (MBP), interferon regulatory factor 5 (IRF5) or gamma T cells. MAP infection prevalence among MS patients is still an unanswered question, although RedHill's MAP companion diagnostic test may help select the right patients for the treatment. Following several proof-of-concept preclinical animal studies, RedHill is conducting a proof-of-concept Phase II trial to investigate whether using RHB-104 in MS patients has any disease-modifying effect.

Phase IIa proof-of-concept ongoing in MS with encouraging interim data

RedHill's [Phase IIa](#) study (CEASE-MS) of RHB-104 in relapsing-remitting multiple sclerosis (RRMS) is ongoing in Israel. RHB-104 is being evaluated as an add-on therapy to IFN-beta1a in an open-label trial of 18 RRMS patients. The primary outcome is combined unique active (CUA) lesions in MR brain scans, while secondary outcome measures include biomarkers, relapse rates, disability status and burden of disease. Although not powered for efficacy, RedHill published encouraging interim [results](#) from this study in March 2016. No clinically significant change was observed for total CUA lesions at week 24, which is supportive of a stable disease state. The annualised relapse rate (ARR, one of the most common primary endpoints in late stage trials in the industry)¹⁵ at 24 weeks was 0.288 in the modified intent-to-treat¹⁶ (mITT) population and 0.0 in the per-protocol¹⁷ (PP) population, comparing

¹³ Burisch et al. The burden of inflammatory bowel disease in Europe. Journal of Crohn's and Colitis, Volume 7, Issue 4, 1 May 2013, Pages 322–337

¹⁴ L. Sechi and C. Dow. Mycobacterium avium ss. paratuberculosis Zoonosis – The Hundred Year War – Beyond Crohn's Disease. Front Immunol. 2015; 6: 96.

¹⁵ A. Lavery et al. Outcome Measures in Relapsing-Remitting Multiple Sclerosis: Capturing Disability and Disease Progression in Clinical Trials. Multiple Sclerosis International. Volume 2014, Article ID 262350.

¹⁶ ITT: the intention-to-treat principle requires that all participants that are randomized must be included in the final analysis regardless of the treatment received, withdrawals, lost to follow-up or cross-overs. This is preferred analysis method to avoid bias. Modified ITT allows some post-randomization exclusions.

¹⁷ Per-protocol analysis is a comparison of treatment groups that includes only those patients who completed the treatment originally allocated. If used alone, this method may lead to bias.

favourably with data published for standalone IFN-beta therapies Avonex (Biogen) 0.67 and Rebif (Merck Serono and Pfizer) 0.87-0.91. A total of 88% of the mITT patient population and 100% of the PP patient population were relapse free at 24 weeks, which also compares well with Rebif (75%) and Avonex (63%). Final results of the 48-week study are expected in Q416.

Market potential in MS

Around 400k people are diagnosed with MS in the US each year alone and around 85% of those have a relapsing-remitting course of the disease, which is in contrast to the progressive type when symptoms gradually get worse over time rather than appearing as relapses. The mainstay of the treatment is disease-modifying therapies with the goal to reduce the frequency of relapses and slowing progression (Medscape) with EvaluatePharma calculating the market will be worth \$25bn in 2020.

Bekinda – bi-modal, extended release ondansetron

Bekinda (formerly RHB-102) is a once-daily, bi-modal release, oral formulation of ondansetron and is being studied at different doses in various gastrointestinal disorders. RedHill in-licensed rights to a technology called CDT (see Exhibit 7 for details) that uses salts to provide an extended release of ondansetron. Bekinda-24mg is in a [Phase III](#) clinical trial in the US to prevent vomiting in acute gastroenteritis/gastritis (the GUARD study), with top-line results expected in mid-2017. Bekinda-12mg recently began [Phase II](#) for diarrhoea-predominant irritable bowel syndrome (IBS-D), with top-line results expected mid-2017.

Ondansetron, originally developed and marketed GlaxoSmithKline and Novartis as Zofran with the patents expired in 2006 and generics available, is a 5-HT₃ (serotonin) receptor antagonist approved for the prevention of chemotherapy- and radiotherapy-induced nausea and vomiting (CINV and RINV, respectively) and prevention of postoperative nausea and/or vomiting (on an as needed basis, not routinely). RedHill licensed the patent-protected extended-release formulation from SCOLR Pharma in May 2010, but now has a direct licence with the Temple University, the original inventor. If approved, it could be the first once-daily oral formulation of ondansetron to reach the US and European markets. RedHill is also pursuing marketing approval of Bekinda in Europe for the prevention of CINV and RINV pending additional feedback from EU member states as to whether additional clinical and preclinical work is necessary.

Gastroenteritis-induced vomiting causes serious morbidity

Acute gastroenteritis is a significant problem, with approximately 179 million cases annually leading to c 600,000 hospitalisations and 5,000 deaths in the US alone ([CDC](#)). Vomiting is very common in children and adolescents with gastroenteritis, leading to dehydration and starvation which can be difficult to treat if the nausea persists. A safe and effective method of controlling vomiting would reduce the need for intravenous rehydration and potentially reduce hospitalisations. A wide range of oral and intravenous anti-emetics are used in adults and children, though concerns have been expressed about side effects of older anti-emetics. Generally, the 5-HT antagonists have few adverse effects and have been safely used in adults and children.¹⁸ Notably one retrospective study of 34,117 paediatric patients found that ondansetron was used in 58% of initial emergency department visits.¹⁹ Rarely ondansetron can be contraindicated in persons who have a congenital long QT syndrome (the time interval between specific waves in the ECG).

¹⁸ Z. Fedorowicz et al. Antiemetics for reducing vomiting related to acute gastroenteritis in children and adolescents. The Cochrane Library, 7 September 2011.

¹⁹ J. Sturm et al. Ondansetron Use in the Pediatric Emergency Department and Effects on Hospitalization and Return Rates: Are We Masking Alternative Diagnoses? Annals of emergency medicine, May 2010, Vol 55.

Phase III GUARD Clinical trial underway in acute gastroenteritis

RedHill began the GUARD [Phase III](#) study in September 2014 to test Bekinda-24mg in 320 patients (aged 12-85 years), with vomiting due to acute gastroenteritis or gastritis. The primary outcome is the proportion of patients without further vomiting, who do not require rescue medication or intravenous hydration in the 24 hours after receiving Bekinda or placebo. The enrolment is expected to be completed in 2017. RedHill believes there is a possibility to file for approval of Bekinda if the GUARD Phase III study is successful, without the need for a second study, but that will depend on further conversations with the regulatory authorities and the data.

Commercial strategy and market potential in gastroenteritis

If approved, Bekinda would be the first 5HT-3 anti-emetic drug indicated for the treatment of acute gastroenteritis. Given that generic ondansetron is already routinely used in this setting, Bekinda's ability to gain market share will depend on its competitive advantage, which is an extended release formulation and will be used on-label compared to standard ondansetron. Ondansetron (orally disintegrating tablets) currently costs around \$30-60 for a two-day course in adults (GoodRx.com). Newer branded treatments that are approved to treat chemotherapy-related nausea, Aloxi (palonosetron iv) and Akynzeo (netupitant/palonosetron) cost around \$500 and \$600/dose respectively (GoodRx.com) with a newer generation 5HT-3 antagonist palonosetron totalling \$456m in sales in 2015 (EvaluatePharma).

Bekinda in IBS-D

IBS is a functional disorder of the bowel that affects around 10-15% of the global population and is characterized by abdominal pain and altered bowel habit (chronic or recurrent diarrhoea, constipation, or both). In the US 25-45 million people are thought to be affected, two-thirds of whom are female, and around half have diarrhoea-predominant IBS (IBS-D) (aboutIBS.org). The causes of IBS are not fully understood but it may result from disturbances in the way the gut, brain and nervous system interact, leading to changes in gut motility.

Some 5-HT-3 receptor antagonists have proved to be effective in IBS by slowing gut transit time. Alosetron (GlaxoSmithKline's Lotronex) is approved for the treatment of women with severe chronic IBS-D but under a restricted prescribing programme due to serious side effects. Ondansetron has a much better safety record and has demonstrated activity in IBS-D in preliminary studies by significantly improving stool consistency, frequency and urgency.²⁰ Garsed et al. conducted a randomized, double-blind, placebo-controlled study with 120 patients with IBS-D and found that half of the usual dose of ondansetron (4mg three times a day instead of 8mg three times a day) led to significantly improved stool consistency and frequency. RedHill believes Bekinda has the potential to be a superior once-daily treatment for patients suffering from IBS-D. In June 2016 RedHill began dosing in a [Phase II](#) trial with Bekinda-12mg for the treatment of IBS-D. The randomized, two-arm parallel group study will be conducted in 16 sites in the US and will enrol 120 patients randomized 60:40 to receive either Bekinda-12 mg or a placebo, once daily, for eight weeks. The primary endpoint for the study is the proportion of patients in each group with response in stool consistency as compared to baseline, per FDA guidance definition. The top-line results are expected mid-2017.

Commercial strategy and market potential in IBS-D

Although studies are at an earlier stage, the opportunity for Bekinda in IBS-D is potentially greater than in gastroenteritis, given the chronic nature of the disorder and the currently underpenetrated market due to a lack of effective drugs. Growth of the US market is likely to be driven by the approval in 2015 of two new drugs for IBS-D, driving both uptake and premium pricing: Salix's Xifaxan (rifaximin, an oral antibiotic), launched mid-2015 and Allergan's Viberzi (eluxadoline, an oral μ -opioid receptor agonist), as a controlled substance, launched slightly later in 2015. Xifaxan had sales of

²⁰ K. Garsed. A randomised trial of ondansetron for the treatment of irritable bowel syndrome with diarrhoea. Gut 2013;0:1-9.

\$220m in the third quarter of 2015, up from \$148m in the previous quarter. Takeda's. Xifaxan's list price is around \$1,800 for 60 tablets, while Linzess costs around \$350 for 30 capsules and Viberzi costs approximately \$1,000 for 60 tablets (GoodRx.com).

Yeliva – first-in-class sphingosine kinase-2 selective inhibitor

Yeliva (formerly ABC294640) is a first-in-class, orally administered sphingosine kinase-2 (SK2) selective inhibitor. Sphingomyelin is not only a building block for cellular membranes but also acts as a precursor for lipid messengers that have profound cellular effects. Sphingosine kinases (there are two isoforms SK1 and SK2) promote rapid production of sphingosine 1-phosphate (S1P). S1P in turn promotes cancer growth, proliferation and inflammation processes. Yeliva is a specific SK2 inhibitor that decreases S1P synthesis, which has been demonstrated to have an effect on a broad range of fundamental biological processes such as cell proliferation, immune cell trafficking and neoangiogenesis. RedHill in-licensed Yeliva from Apogee Biotechnology on 31 March 2015 (see Exhibit 7). Apogee has demonstrated versatile effects of Yeliva in a number of preclinical studies. Acute and chronic animal toxicology studies showed excellent toxicity profile with a dose of 100mg/kg, at least 10-times below the lowest observed adverse effect level, resulting in plasma level above the IC₅₀ of human SK2, which is sufficient for suppression of tumour cell proliferation.²²

First Phase I study endpoints met, development plan established

Apogee has also completed an open-label, dose-escalation, first-in-human [Phase I](#) trial with Yeliva in 21 patients with solid tumours, including gastrointestinal cancers such as pancreatic, colorectal cancers and cholangiocarcinoma. Key findings include good safety profile, drug-like PK/PD profile and first-ever longitudinal analysis of plasma S1P levels as a potential biomarker. Of the 16 patients one had a partial response and six patients had stable disease. Based on these findings RedHill prepared a development plan for Yeliva to explore it in promising indications (Exhibit 5).

Exhibit 5: Development plan for Yeliva

Study	Description	N	Status	Endpoints
Phase I/II NCT02229981	Safety and preliminary efficacy study in refractory or relapsed DLBCL , including patients with virus-induced (eg Kaposi's sarcoma-associated herpesvirus or Epstein-Barr virus) lymphoma.	Up to 33	Initiated June 2015 On administrative hold due to protocol amendments to improve recruitment	Primary: MTD, DLT Secondary: PK/PD, plasma tumour progression radiographic assessment (PET criteria)
Phase I/II NCT02757326	Safety, open-label, dose escalation and efficacy study in patients with refractory or relapsed multiple myeloma that have previously been treated with proteasome inhibitors and immunomodulatory drugs.	Up to 77	Initiated in September 2016	Primary: MTD (Ib); overall response rate and OS (II); Secondary: PK/PD, pharmacodynamic markers
Phase II NCT02939807	A Phase II study to evaluate the efficacy and safety of Yeliva as a second-line monotherapy for patients with advanced hepatocellular carcinoma relapsed after treatment with first-line single-agent sorafenib (Nexavar).	Up to 39	Initiated in October 2016	Primary: objective response rate (modified RECIST) Secondary: TTP, OS, PK/PD, pharmacodynamic markers, adverse events
Phase Ib	Safety and efficacy study of Yeliva in the prevention of mucositis from radiotherapy for treatment of head and neck carcinoma.	TBD	TBD	TBD

Source: RedHill, Edison Investment Research. Note: DLBCL – diffused large B-cell lymphoma; MTD - maximum tolerated dose; DLT - dose limiting toxicity; OS - overall survival; RECIST - Response Evaluation Criteria in Solid Tumours; TTP - time to tumour progression

Competitive landscape: Mechanism of action

Historically the sphingosine-1-phosphate receptor (target for S1P) modulation was first explored for therapeutic potential. Yeliva acts more upstream and decreases S1P (antagonist effect), while modulators can act as agonists or functional antagonists depending on the tissue or cell context. Fingolimod (Gilenya, Novartis) was the first orally available, small molecule S1P receptor modulator that was shown to have a disease modifying effect by reducing the rate of relapses in multiple sclerosis patients by approximately one-half over a two-year period. Global 2015 sales were \$2.8bn (EvaluatePharma). With patents due to start expiring in 2019, Novartis has a follow-on programme, siponimod, with a similar mode of action in Phase III. Yeliva has a differentiated mechanism of action

and another potential competitive advantage is the well-known fact that fingolimod and likely other S1P modulators cause a dose-dependent decrease in white blood cells, which can compromise the immune system. RedHill noted that Yeliva's effect on white blood cells is modest.

Rizaport – new API delivery option for migraineurs

Rizaport is an oral dissolving thin film formulation of rizatriptan for the treatment of migraine attacks. Migraine is a complex neurological disorder defined by recurrent episodes of characteristic headaches, which in some cases last for two to three days and incapacitate the patient. While mild migraine can be treated with simple analgesics, moderate to severe cases rely on migraine-specific triptan class drugs with several other types of medication available as well. Triptans act as agonist on serotonin 5-HT_{1B} and 5-HT_{1D} receptors in painfully dilated cranial blood vessels causing them to constrict and in nerve endings inhibiting the release of vasoactive neuropeptides.²¹ Triptans are an established class of drugs first introduced in 1990s with a variety of active pharmaceutical ingredients (APIs) off patent. The popularity of this class is mainly due to its high efficacy for classic moderate migraine attacks. For example, sumatriptan, one of the first drugs in this class, demonstrated 67-79% headache reduction rate four hours after the oral administration. Rizatriptan has similar efficacy, but potentially is one of the fastest acting triptans. Rizatriptan was originally developed by Merck & Co and marketed as Maxalt (oral and Maxalt-MLT orally disintegrating tablets) after the FDA approval in June 1998. It had reached peak sales of c \$640m in 2011 before patents expired in 2012.

RedHill and IntelGenx collaboration on Rizaport

IntelGenx, a Canada-based oral drug delivery formulation company, developed the orally dissolving, thin film formulation VersaFilm. RedHill and IntelGenx entered into a co-development agreement for Rizaport in August 2010. The two companies jointly finalised the development (Phase I bioequivalence vs Maxalt-MLT) in 2012. In March 2013 the two companies submitted an NDA to the FDA under 505(b)(2) regulatory path, but received a complete response letter primarily focusing on third-party chemistry and CMC issues, packaging and labelling. The FDA did not question safety or clinical trial results, therefore RedHill plans to address the issues and resubmit the NDA in H117. In Europe, the two partners received approval for Marketing Authorisation Application. On 5 July 2016, RedHill announced the first commercialisation agreement with Grupo Juste in Spain with potential expansion into other territories in Latin America and Middle East. Financial details were undisclosed with the launch in Spain expected to take place in H217. On 21 September 2016, RedHill announced a binding term sheet with the second commercialization partner Pharmatronic for the commercialisation of Rizaport in South Korea, with the launch seen in Q119. The main competitive advantage of the orally dissolving formulation compared to conventional tablets is evident in the case of headaches with high-frequency nausea (ie, more than half the time), which also accompanies migraine attacks in around 50% of cases, with milder nausea being even more prevalent.²² An injectable formulation requires a healthcare setting, while the patch formulation is a novel approach with the first product (Zecuity, Teva) approved just in 2013, but sales were temporarily suspended in June 2016 after suspicion of burns and scars associated with its use.

Sensitivities

RedHill is subject to the usual risks associated with drug development, including clinical development failures, regulatory risks, competition, partnering setbacks and financing and commercial risks. Some of the company's lead products (eg RHB-105, RHB-104 and Bekinda) are repositioned or reformulated generic drugs, which implies a lower R&D risk. On the other hand, if approved they may

²¹ Tepper, S. J.; Rapoport, A. M.; Sheftell, F. D. (2002). "Mechanisms of action of the 5-HT_{1B/1D} receptor agonists". *Archives of neurology*. 59 (7): 1084–1088.

²² R. Lipton. Frequency and burden of headache-related nausea: results from the American Migraine Prevalence and Prevention (AMPP) study. *Headache*. 2013 Jan;53(1):93-103.

face generic substitution risk. The competitive advantage of these products versus cheaper generics will dictate the uptake of the market. Also, the composition of matter patents have already expired for the aforementioned products, making the overall IP protection more vulnerable to challenges; although formulation patents may provide adequate protection. The biggest near-term sensitivities are related to RHB-105, RHB-104 and Bekinda, which all are in late-stage development. Besides R&D risk, there is also substantial uncertainty about the marketing of the products as they all target rather large patient populations treated in different healthcare settings; therefore, strong marketing partners are likely to be needed. This, however, might not be an issue depending on the data obtained in the clinical trials. As discussed, all the products have competitive advantages over generic drugs, but the eventual market uptake is difficult to forecast. On 1 November 2016 RedHill announced a public offering, which was subsequently withdrawn due to market conditions. The amount of funds the company was seeking was not revealed.

Valuation

We value RedHill based on risk-adjusted NPV analysis using a 12.5% discount rate and the 52-week NIS exchange rate range to the US dollar (3.74-3.99), including \$40.5m net cash at end Q316. This corresponds to a value of NIS1.21-1.29bn or NIS9.5-10.1/share. Exhibits 6 and 7 provide assumptions and our valuation of assets in a specific indication. We included five of the company's eight assets in a total of nine indications. We see most of RedHill's value in its mid- to late-stage products, excluding programmes in preclinical stages – Mesupron and RP101 – although they both have been explored in clinical trials before and if RedHill obtains convincing preclinical data, we believe the transition into the clinic could happen rapidly and would provide upside to our valuation. We also excluded Bekinda's indication for oncology support, as we await clarification of whether additional development will be needed in Europe and the extent of the supportive studies needed in the US. Lastly, we do not yet include RHB-106 in our valuation, although we see RHB-106 as clearly differentiated product. This is mainly due to change in the ownership of the rights to the asset after Valeant acquired Salix more than a year ago but has yet to clarify further development plans.

Exhibit 7 summarises the rest of our assumptions for RedHill's valuation. For the clinical projects we have used standard industry assumptions. Probabilities to reach the market and timelines were selected according to the stage of the project. Product launch dates were estimated based on necessary additional development. RedHill's strategy is to develop the products standalone or out-license in later stages. Only RHB-106 has been out-licensed so far, while other products are still wholly owned. We assume that the development of the products in Phase III will be finalised by RedHill, but a commercial partner or distributor will be sought due to the number of specialty areas in which the company is involved (5% COGS margin and 30% sales and marketing margin assumed, with the rest retained by RedHill). For the calculation of target patient groups, we use the US population plus the top five European countries, Benelux, the Nordics and Austria with Switzerland.

Exhibit 6: Sum-of-the parts RedHill valuation

Product	Launch	Peak sales, \$m	NPV, (\$m)	NPV/share, (\$)	Probability	rNPV (\$m)	rNPV/share (\$)
RHB-105, - <i>H. pylori</i> infection	2021	86	80.1	0.6	70%	53.2	0.4
RHB-104, - Crohn's disease	2023	145	44.8	0.4	40%	7.5	0.1
- Multiple sclerosis	2025	422	179.1	1.4	20%	45.4	0.4
Bekinda, - Gastroenteritis	2019	21	30.5	0.2	70%	20.3	0.2
- IBS-D	2023	201	110.8	0.9	40%	58.1	0.5
Yeliva, - r/r MM	2025	565	212.1	1.7	10%	42.4	0.3
- Advanced HCC	2025	649	119.4	0.9	10%	29.2	0.2
- DLBCL	2025	156	60.6	0.5	10%	15.6	0.1
Rizaport, - Migraine	Market	20	10.6	0.1	100%	10.6	0.1
Net cash (including other financial assets)			40.5	40.5		100%	40.5
Valuation			888.4	6.7		322.8	2.5

Source: Edison Investment Research. Note: WACC = 12.5% for product valuations. IBS-D: irritable bowel syndrome; r/r MM: refractory/relapse multiple myeloma; Advanced HCC: hepatocellular carcinoma; Diffuse large B-Cell Lymphoma.

Financials

RedHill reports in US dollars, while the expenses are mostly denominated in US dollars and Israeli shekels. RedHill had cash and cash equivalents (including bank deposits and financial assets at fair value) of \$40.5m at the end of Q316 compared to \$64.2m in Q315 and was debt free. In line with increasing R&D activities, RedHill's net cash used in operations increased from \$3.7m in Q315 to \$7.4m in Q316, the majority of which is attributable to R&D. R&D expenditure totalled \$7.0m in Q316 (\$3.9m in Q315). Q316 G&A costs amounted to \$1.4m (\$0.7m in Q315). Our total R&D and G&A forecasts for 2016 are \$20.4m and \$4.8m respectively. As a result, we expect a cash position of \$33.2m by the end of this year. RedHill does not provide guidance, but according to our model and based on current R&D plans the cash reach is into Q417, past several key R&D catalysts. Our estimated need for additional funds in 2018 is around \$28m, which we include as illustrative long-term debt in our financial forecasts, although the company may seek to raise this sooner. Notably, we do not take into account potential revenues from Rizaport or other products and we do not include any potential licensing-related income in our financial forecasts. We only included the received grants in our model and due to low visibility we do not forecast any additional grants although RedHill may manage to attract additional non-dilutive funding.

Exhibit 7: Assumptions for R&D and commercial projects

Product / stage / indication	Out-licensing assumptions	Comments
RHB-105 rifabutin+ amoxicillin+ omeprazole - Ph III - <i>H. pylori</i>	Develop standalone	- <u>Target population</u>: c 640k: rescue therapy for patients with dyspepsia who are treated for <i>H. pylori</i> but failed two or three times. Assumed 30% penetration. Potential for increase if RHB-105 in front-line treatment. - <u>Pricing*</u>: \$400 per treatment course, with a discount to branded a/b combos; peak sales in seven years. - <u>R&D cost</u>: \$17.5m for second Phase III. - <u>Rights</u>: purchase agreement with Giaconda for RHB-104, RHB-105 and RHB-106 in August 2010 at a price of \$500,000, 7% royalties from net sales or 20% of the royalties from sublicensees. Last patent expires in 2034.
RHB-104 clarithromycin+ clofazimine+ rifabutin - Ph III - Crohn's disease	Develop standalone	- <u>Target population</u> of c 308k: 50% of new cases plus 35% of existing patients who are expected to relapse within one year, 50% due to sig. variability in MAP infection in CD. Assumed 20% penetration, conservative as the treatment can be highly complementary. - <u>Pricing*</u>: \$2,000 on average per treatment. Pricing per two-week course same as in <i>H. Pylori</i> infection. Assumed 12-week treatment on average and a discount to avoid generic substitution. Peak sales reached in seven years. - <u>R&D cost</u>: \$12m for the remaining first Phase III with data in H218/H119; \$15m for second Phase III afterwards - <u>Rights</u>: see RHB-105. Last patent expiry date 2029.
- Ph IIa - r/r multiple sclerosis	Licensing deal in 2020 after Ph IIb; \$36m upfront, \$441m in milestones (average of five deals in MS area over past five years)	- <u>Target population</u> of c 376k: r/r MS is 85% of the total prevalence; assumed 50% MAP infection rate, but the data is virtually non-existent. Assumed 20% penetration, conservative as the treatment can be highly complementary. - <u>Pricing*</u>: \$4,000 on average per treatment. Pricing per two-week course same as in <i>H. Pylori</i> infection. Assumed 24-week treatment on average per patients and a discount to avoid generic substitution. Peak sales reached in seven years. - <u>R&D cost estimate</u>: \$10m for Phase IIb (Phase IIa full top-line results in Q416), then out-licensed. - <u>Rights</u>: see RHB-105. Last patent expiry date 2034.
Bekinda ext. release tab. ondansetron - Ph III - Gastroenteritis	Develop standalone	- <u>Target population</u> of c 663k: around half of the gastroenteritis patients admitted to emergency departments already are treated with 5-HT₃ receptor antagonists. Assumed 20% penetration due to highly fragmented market with branded and generic drugs. - <u>Pricing*</u>: \$135 on average per three-day treatment, implies 50% premium to generic orally disintegrating ondansetron. Peak sales reached in seven years. - <u>R&D cost</u>: \$3.6m for the remainder of Phase III (enrolment complete in early 2017), likely sufficient for the NDA. - <u>Rights</u>: in-licensed from Temple University in March 2014. Details undisclosed. Last patent expiry date 2035.
- Ph II - IBS-D	Licensing deal in 2018 after the ongoing Ph II; \$53m upfront, \$249m in milestones (average of two deals in IBS area over past five years)	- <u>Target population</u> of c 1.2 million: population prevalence of around 11%; around 30% of all patients will consult a doctor, around half will have IBS-D. Assumed that severe patients (10%) will be most accessible to Bekinda. Assumed 10% penetration due to highly fragmented market with branded and generic drugs. - <u>Pricing*</u>: \$1,260 on average per eight-week treatment, implies 50% premium to generic orally disintegrating ondansetron (only half dose as in gastritis case). Peak sales reached in seven years. - <u>R&D cost estimate</u>: \$4.4m for remainder Phase II (full top-line results in mid-2017), then out-licensed - <u>Rights</u>: see above for licensing information. Last patent expiry date 2035
Yeliva sphingosine kinase-2 inhibitor - Ph I/II - r/r multiple myeloma	Licensing deal in 2020 after the ongoing Ph I/II; \$57m upfront, \$621m in milestones (average of seven deals in MM area over past five years)	- <u>Target population</u> of c 67k: new cases of MM. As MM is not curable the disease will ultimately relapse. Assumed 20% market penetration due to unproven mechanism and availability of several novel drugs, but MM is still an unmet need. - <u>Pricing*</u>: assumption is that Yeliva will have to be priced similarly for all three indications. Bottom up analysis shows that \$30k is reasonable and similar in MM and HCC, but higher than DLBCL (due to threat of biosimilars). Peak sales reached in six years. - <u>R&D cost estimate</u>: \$10.1m for upcoming Phase I/II, of which \$2m grant from the NCI; then out-licensed - <u>Rights</u>: in-license agreement with Apogee for an upfront of \$1.5m, \$4m in milestones (\$2m recognized as liability), tiered royalties starting in low double digits. Last patent expiry date 2031.
- Ph II - Advanced HCC	Licensing deal in 2020 after the ongoing Ph II; \$54m upfront, \$376m in milestones (average of four deals in HCC area over past five years)	- <u>Target population</u> of c 51k: new cases that are not eligible for curative therapy (around 30%) such as surgery. Assumed 30% penetration as still highly unmet need with few options. - <u>Pricing*</u>: see above - <u>R&D cost</u>: \$8.1m for upcoming Phase II, of which a portion of \$1.8m grant from the NCI to the investigator; then out-licensed - <u>Rights</u>: see above for licensing information. Last patent expiry date 2031
- Ph I/II - DLBCL	Licensing deal in 2020 after the ongoing Ph I/II; \$33m upfront, \$162m in milestones (one third of the average of five deals in DLBCL area – due to biosimilar competition)	- <u>Target population</u> of c 19m: 30-40% of new cases, of which one third will be r/r DLBCL. Assumed 20% penetration due to upcoming number of rituximab biosimilars. Also Yeliva's pricing might be prohibitory for more significant uptake. On the other hand, if Yeliva proves superiority over rituximab or useful in combination, then the penetration might be significantly higher. - <u>Pricing*</u>: see above - <u>R&D cost estimate</u>: \$4.3m for upcoming Phase I/II, incl. an undisclosed grant from the NCI; then out-licensed. - <u>Rights</u>: see above for licensing information. Last patent expiry date 2031
Rizaport oral thin film - rizatriptan - Market Migraine	Partnership with IntelGenx ¹ marketed in Europe; NDA re-submission expected next year. We assume Rizaport will be marketed via distributors and use 50% margin from net sales to account for this.	- <u>Target population</u> includes classic migraine. Due to fragmented market we employed a top-down approach and used Maxalt MLT as a peer drug. After the patent expiry in 2012 sales of Maxalt MLT fell from \$334m to \$52m in the US in 2013. Sales in 2015 were \$16m with a slight down trend. We use \$20m in peak sales for Rizaport in the US and top European countries and assume peak sales reached slowly over 10 years. 10% COGS margin, 50% S&M margin and royalties as below. - <u>Rights</u>: joint development and commercialisation agreement with IntelGenx in August 2010. IntelGenx retained manufacturing rights. RedHill paid \$800k in upfront and milestones so far + \$500k upon the FDA approval. 20% royalties to IntelGenx if RedHill markets itself or 60% of the first \$2m sublicense fees and 40% thereafter if marketed by sublicensees. Financial details from the agreement with Grupo Juste for distribution in Spain remain undisclosed. Last patent expiry date for thin film formulation until around 2035.

Source: Edison Investment Research, RedHill. Note: IBS-D: irritable bowel disease with diarrhoea; DLBCL: diffuse large B-cell lymphoma; HCC: hepatocellular carcinoma; *pricing in US; 20% discount applied in Europe; licensing deal source EvaluatePharma.

Exhibit 8: Financial summary

	\$'000s	2010	2011	2012	2013	2014	2015	2016e	2017e
December		IFRS	IFRS	IFRS	IFRS	IFRS	IFRS	IFRS	IFRS
PROFIT & LOSS									
Revenue		0	23	16	12	7,014	3	0	100
Cost of Sales		0	0	0	0	0	0	0	(60)
Gross Profit		0	23	16	12	7,014	3	0	40
Research and development		(736)	(5,414)	(6,455)	(8,100)	(12,700)	(17,771)	(20,437)	(28,014)
EBITDA		(1,733)	(7,858)	(9,016)	(10,748)	(10,620)	(21,966)	(25,200)	(32,976)
Operating Profit (before amort. and except.)		(1,733)	(7,873)	(9,040)	(10,772)	(10,647)	(22,002)	(25,237)	(33,014)
Intangible Amortisation		0	0	0	0	0	0	0	0
Exceptionals		0	0	0	0	0	0	0	0
Other		0	0	0	0	0	0	0	0
Operating Profit		(1,733)	(7,873)	(9,040)	(10,772)	(10,647)	(22,002)	(25,237)	(33,014)
Net Interest		(811)	(7,630)	(1,286)	144	(64)	912	187	109
Profit Before Tax (norm)		(2,544)	(15,503)	(10,326)	(10,628)	(10,711)	(21,090)	(25,050)	(32,904)
Profit Before Tax (reported)		(2,544)	(15,503)	(10,326)	(10,628)	(10,711)	(21,090)	(25,050)	(32,904)
Tax		0	0	0	0	0	0	0	0
Profit After Tax (norm)		(2,544)	(15,503)	(10,326)	(10,628)	(10,711)	(21,090)	(25,050)	(32,904)
Profit After Tax (reported)		(2,544)	(15,503)	(10,326)	(10,628)	(10,711)	(21,090)	(25,050)	(32,904)
Average Number of Shares Outstanding (m)		0.0	48.1	52.6	62.4	86.6	110.8	127.3	127.5
EPS - normalised (\$)		(265.00)	(0.32)	(0.20)	(0.17)	(0.12)	(0.19)	(0.20)	(0.26)
EPS - normalised and fully diluted (\$)		(265.00)	(0.32)	(0.20)	(0.17)	(0.12)	(0.19)	(0.20)	(0.26)
EPS - (reported) (\$)		(265.00)	(0.32)	(0.20)	(0.17)	(0.12)	(0.19)	(0.20)	(0.26)
Dividend per share (\$)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Gross Margin (%)		N/A	100.0	100.0	100.0	100.0	100.0	N/A	40.0
EBITDA Margin (%)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Operating Margin (before GW and except.) (%)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
BALANCE SHEET									
Fixed Assets		1,208	1,450	1,533	1,739	2,837	6,318	7,281	8,249
Intangible Assets		1,200	1,245	1,345	1,555	2,615	6,060	7,060	8,060
Tangible Assets		8	132	113	103	146	124	87	55
Investments		0	73	75	81	76	134	134	134
Current Assets		9,215	17,172	17,498	12,358	26,019	60,510	35,205	2,935
Stocks		0	0	0	0	0	0	0	0
Debtors		150	89	198	488	3,074	2,372	1,978	1,978
Cash		9,050	14,070	16,814	11,851	5,892	21,516	14,916	957
Other		15	3,013	486	19	17,053	36,622*	18,311*	0
Current Liabilities		(141)	(513)	(1,078)	(2,415)	(1,720)	(5,514)	(4,664)	(4,707)
Creditors		(141)	(513)	(1,078)	(2,415)	(1,720)	(5,514)	(4,664)	(4,707)
Short term borrowings		0	0	0	0	0	0	0	0
Long Term Liabilities		(11,337)	(886)	0	0	0	0	0	0
Long term borrowings		(10,619)	0	0	0	0	0	0	0
Other long term liabilities		(718)	(886)	0	0	0	0	0	0
Net Assets		(1,055)	17,223	17,953	11,682	27,136	61,314	37,822	6,476
CASH FLOW									
Operating Cash Flow		(1,632)	(4,691)	(6,795)	(8,436)	(12,229)	(17,826)	(23,911)	(31,266)
Net Interest		0	0	0	0	0	0	0	0
Tax		0	0	0	0	0	0	0	0
Capex		(0)	(136)	(8)	(14)	(70)	(14)	0	(5)
Acquisitions/disposals		0	0	0	0	0	0	0	0
Financing		0	13,838	6,550	2,280	24,369	54,792	0	0
Other		566	6,628	2,997	1,207	(18,029)	(21,328)	17,311**	17,311**
Dividends		0	0	0	0	0	0	0	0
Net Cash Flow		(1,066)	15,639	2,744	(4,963)	(5,959)	15,624	(6,600)	(13,960)
Opening net debt/(cash)		(597)	1,569	(14,070)	(16,814)	(11,851)	(5,892)	(21,516)	(14,916)
HP finance leases initiated		0	0	0	0	0	0	0	0
Other		0	0	0	0	0	0	0	0
Closing net debt/(cash)		469	(14,070)	(16,814)	(11,851)	(5,892)	(21,516)	(14,916)	(957)

Source: Edison Investment Research, RedHill accounts. Note: *Bank deposits and financial assets at fair value. **Includes bank deposits converted to cash and cash equivalents.

Contact details	Revenue by geography
RedHill Biopharma Ltd. 21 Ha'arba'a St., Tel-Aviv 6473921, Israel +972-(0)3-541-3131 http://www.redhillbio.com	N/A
Management team	
Chairman and chief executive officer: Dror Ben-Asher Mr Ben-Asher co-founded Redhill in 2009 and has been its CEO since then. Prior to this, he was President and CEO of ProSeed Capital Holdings, a European Venture Capital fund which he co-founded. Mr Ben-Asher also served as a consultant for a leading Israeli law firm and the Harvard International Law Journal at Harvard Law School.	Chief financial officer: Micha Ben Chorin Mr Chorin joined Redhill in 2016 having previously served as CFO (and Executive President) of WiNetworks and Interlogic, prior to that he was the CFO of Global Village Telecom (GVT) in Brazil and Corporate Controller of GVT in the US.
Chief business officer: Guy Goldberg Prior to joining RedHill, Mr Goldberg served as Senior VP of Business Operations at Eagle Pharmaceuticals, a specialty injectable drug development company. Mr Goldberg's previously also worked at a healthcare focused venture capital firm ProQuest Investments and as a consultant at McKinsey & Co.	Chief operating officer: Gilead Raday Mr Raday previously served as interim CEO of Sepal Pharma and as a director at TK Signal and Morria Biopharmaceuticals. He is a graduate from the University of Cambridge (Bioscience Enterprise) and the Hebrew University of Jerusalem (Neurobiology, Mathematics and Biology).
Principal shareholders	(%)
Broadfin Capital	5.78
Orbimed Advisors	4.43
Menora Mivtachim Holdings	1.93
Sabby Management	1.16
Candriam Investors Group	0.56
Ayalon Asset Management	0.41
Bank Hapoalim	0.36
Companies named in this report	
Pfizer (PFE), Gilaconda, Pharmacia, Takeda (4502), Allergan (AGN)	

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